NeuroSTEM

Computing of Neuroscience Data

Companion quiz

1. What is one example of what MRI is useful for?

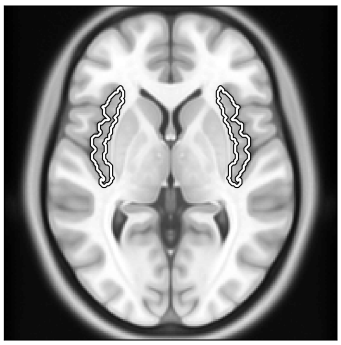
2. What is one example of what PET is useful for?

3. Name the brain area outlined in each picture.

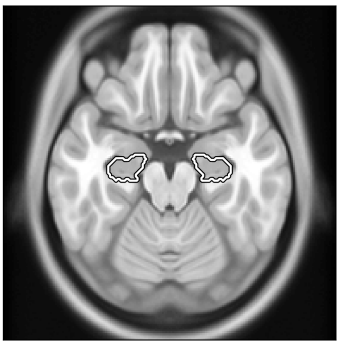
The pictures show different slice levels through the brain.

Which brain area is outlined?

Picture A



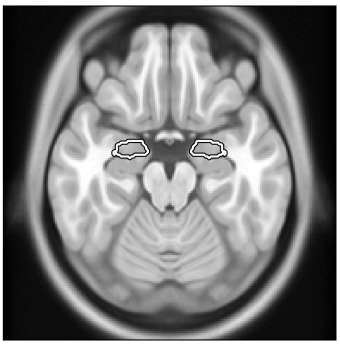
Picture B



Picture C



Picture D



A: _____ **B:** _____

C: _____ **D:** _____

4. Why do we line up (register) the MRI and PET images before measuring PET signal in a brain area?

5. Is a relationship between stress and PET signal suggested by the data?

For each group size, write yes, no, or unsure for each region.

Participants	Hippocampus	Amygdala	Insula	Anterior cingulate
5				
10				
50				
500				

Workshop: https://ashgillman.github.io/neurostem-2026/

Teaching examples: workshop PET signals and participant data are simulated.

MRI: MNI152 (Fonov et al., 2011); atlas: CerebrA (Manera et al., 2020), via TemplateFlow.

Worksheet: Ashley Gillman, CC BY 4.0. Image terms and notices: DATA_SOURCES.md / NOTICE.

MRI template: Copyright (C) 1993-2004 Louis Collins, McConnell Brain Imaging Centre,
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